

Advanced Data Science

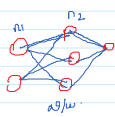
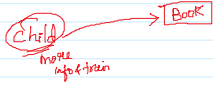
Machine Learning

- Subset of AI
- Structured data (table)
- Small dataset
- Computing time is less
- CPU is sufficient to train the model in fast execution
- Kmeans, SVM, DT, etc.

Deep Learning

- Subset of ML
- Structured & unstructured (image, audio, video)
- Large Ab. of dataset
- Computing time is more
- GPU is not sufficient.
- Several GPUs → $2-3$ → 34

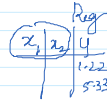
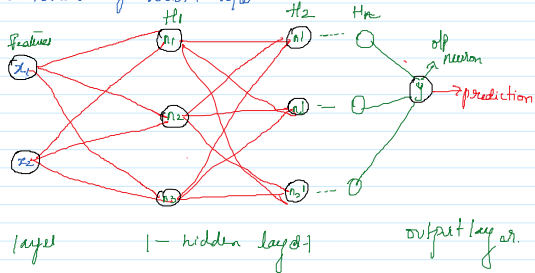
How human brain works?



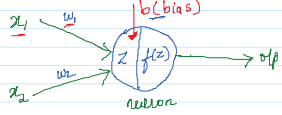
Artificial brain →

Neurons
↓
Neural net.

Architecture of Neural Net.



How neuron works?



$$z = x_1 w_1 + x_2 w_2 + b$$

$f(z) \rightarrow$ function of z

① $f(z) \Rightarrow$ Relu function

$$f(z) = \max(z, 0)$$

Assume

Initial the model with select Random values of $w_1 + b$
 $x_1 = 4$ $x_2 = 5$
 $w_1 = 2$ $w_2 = 3$
 $b = 1$

$$z = x_1 w_1 + x_2 w_2 + b$$

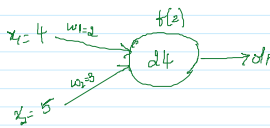
$$z = 24$$

$$f(z) = \max(z, 0)$$

$$\max(24, 0) = 24$$

Eg: $\max(25, 0)$
↓
 25
② $\max(-3, 0)$
↓
 0

Relu → filter the Value



Output layer

Regression

Linear Activation function

only 1 neuron

Bi-classifi

① → class 0 → No

② → class 1 → Yes

$f(z) \Rightarrow$ Sigmoid Activation function

df Prob

0

1

C (1/5 - 3 labels)

Multi-classi

③ → Softmax

④ → max

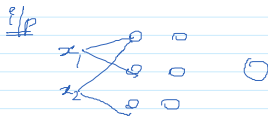
⑤ → Negative

$f(z) \Rightarrow$ Softmax

$f(z) \Rightarrow$ Softmax
Activation
func.

hidden layer $\Rightarrow f(z) \Rightarrow$ ReLU

x_1, x_2, y



if rows $\Rightarrow$ $\begin{matrix} x_1 & x_2 \\ \hline 2 \text{ rows} \\ B_1 \end{matrix}$ $\begin{matrix} x_3 & x_4 \\ \hline 2 \text{ rows} \\ B_2 \end{matrix}$

Regularization: Technique used to Reduce Overfitting

Conduct $\therefore Q_1, Q_2, Q_3 \dots Q_{10}$ of same $\rightarrow$

I $\frac{10}{10} \Rightarrow$ Very good $\rightarrow$ Questions are known to shalini

If I Conduct test and Questions are new

I $\frac{2}{10} \Rightarrow$ Very bad/not good $\Rightarrow$ New Question

$\therefore$ Shalini performance is very good for known Question but failed to perform for New/unknown Question.

epoch $\rightarrow$ one time
trained
the
data
2 epoch
1
n epoch

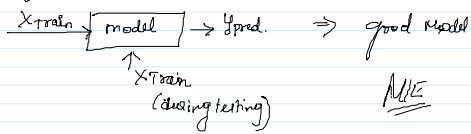
20 seconds
 $\begin{matrix} x_1 & x_2 & x_3 & x_4 & x_5 \\ \hline B_1 \end{matrix}$ $\begin{matrix} x_6 & x_7 \\ \hline B_2 \end{matrix}$ $\begin{matrix} x_8 & x_9 & x_{10} \\ \hline B_3 \end{matrix}$ $\begin{matrix} x_{11} & x_{12} & x_{13} & x_{14} & x_{15} \\ \hline B_4 \end{matrix}$

random rows
 $\begin{matrix} x_{16} & x_{17} & x_{18} & x_{19} & x_{20} \\ \hline B_1 \end{matrix}$ $\begin{matrix} x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \\ \hline B_2 \end{matrix}$ B_3 B_4

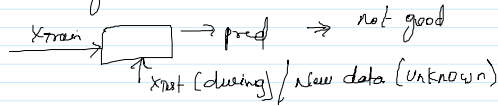
1 epoch
how many
times I
have
trained entire
data
2nd time
2 epoch

Model Works very well for the Trained data.

Model works very well for the trained data.



but failed to work for test data.



Regularization: To reduce overfitting

Shrink I $\frac{10}{10} = \text{good}$.

II $\frac{3}{10} \Rightarrow \frac{4}{10} \quad \frac{3}{10} \quad \frac{4}{10}$

by adding the penalty factor.

I L_1 (Lasso)

$$\text{Total loss} = \text{Error} + \text{penalty factor} \\ (n-p) \text{ MSE} + \lambda \sum_{i=1}^n w_i$$

$\therefore$ if any features have less correlation with $y \Rightarrow$ remove the feature

II L_2 (Ridge)

$$\text{Total loss} = \text{Error} + \lambda \sum_{i=1}^n w_i^2$$

$\therefore$ F_1, F_2, F_3, F_4 Target
(20%) (80%) (10%) (10%)
 F_2, F_4

Randomly model selected

w_1, w_2, b

batch 1 data

N/w

pred

error

error + Penalty $\Rightarrow$ gradient

$w_{\text{new}} = w_{\text{old}} - \eta \text{ gradient}$

$w_{\text{new}}, w_{\text{new}}, b_{\text{new}}$

find the updated values of w & b

batch 2

$w_1, w_2, b \Rightarrow$ updated weights

$\downarrow$ train the model

$\downarrow$ weight $\rightarrow$ batch 1

↓ train the "
↓ updated weight → broken